

Comparison of Numerical Methods for Solving Initial Value Problems in MATLAB

Numerical Methods Portfolio Project

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MATLAB Online

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Methods: Euler Method, Taylor Method of Order n (2,3), Midpoint Method, Modified Euler Method, and Runge-Kutte Order 4 Method

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1 PROJECT OVERVIEW

Numerical methods for Initial Value Problems (IVPs) are widely used in engineering applications where analytical solutions are difficult or impossible to obtain. These methods provide approximate solutions to ordinary differential equations and form the foundation of computational techniques used in dynamics, control systems, heat transfer, fluid mechanics, and robotics.

This project implements and compares five numerical methods for solving a first-order ordinary differential equation in MATLAB. The methods are evaluated based on accuracy, convergence characteristics, and computational performance. A convergence study is also performed to investigate the effect of step size on solution accuracy.

2 PROBLEM STATEMENT

The following initial value problem was selected:

$$\frac{dy}{dt} = te^{3t} - 2y$$

with the initial condition:

$$y(0) = 0$$

over the interval:

$$0 \leq t \leq 1$$

using a step size of:

$$h = 0.1$$

The numerical solution obtained from each method was compared with the exact solution.

3 EXACT SOLUTION

The differential equation is a first-order linear ordinary differential equation. Using the integrating factor method, the exact solution is:

$$y(t) = \frac{1}{25} [(5t - 1)e^{3t} + e^{-2t}]$$

This exact solution serves as the benchmark for evaluating the accuracy of the numerical methods.

4 METHODS IMPLEMENTED

Five numerical methods were implemented:

4.1 EULER METHOD

The Euler Method uses the slope at the current point to estimate the next value of the solution.

4.2 TAYLOR METHOD (ORDER 2 AND 3)

The Taylor Method improves accuracy by including higher-order derivative information in the approximation.

4.3 MIDPOINT METHOD

The Midpoint Method evaluates the slope at the midpoint of each interval to obtain a more accurate estimate than Euler's Method.

4.4 MODIFIED EULER METHOD

The Modified Euler Method uses a predictor-corrector approach by averaging slopes at the beginning and end of the interval.

4.5 RUNGE-KUTTA FOURTH ORDER METHOD (RK4)

The RK4 Method uses four slope evaluations per step and is generally considered one of the most accurate fixed-step numerical methods.

5 MATLAB IMPLEMENTATION

Each method was implemented as a separate MATLAB script. Every script calculated:

- Numerical solution values
- Exact solution values
- Absolute errors
- Graphical comparison with the exact solution

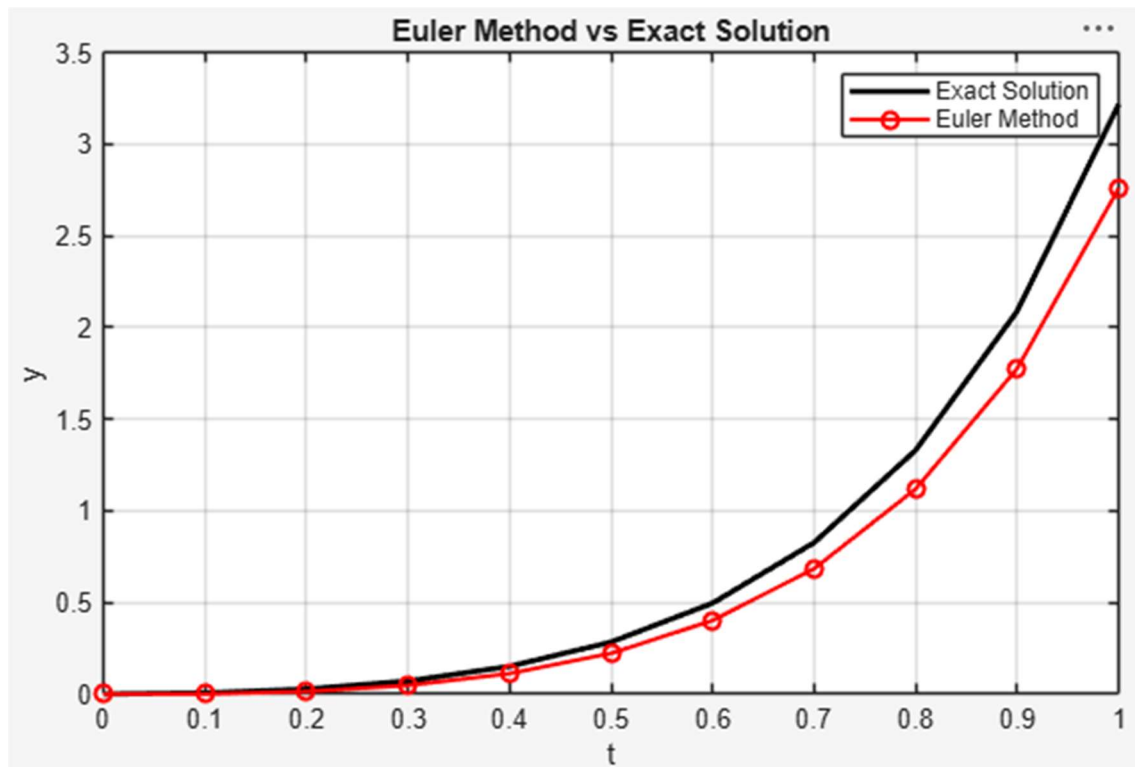
A comparison script was also developed to evaluate the performance of all methods simultaneously and generate comparison plots.

6 INDIVIDUAL METHOD RESULTS

6.1 EULER METHOD

The Euler Method produced a reasonable approximation but showed noticeable deviation from the exact solution as the solution progressed.

t	y	y_exact	error
0.0	0.000000	0.000000	0.000000
0.1	0.000000	0.005752	0.005752
0.2	0.013499	0.026813	0.013314
0.3	0.047241	0.071145	0.023903
0.4	0.111581	0.150778	0.039197
0.5	0.222070	0.283617	0.061547
0.6	0.401740	0.496020	0.094279
0.7	0.684371	0.826481	0.142110
0.8	1.119129	1.330857	0.211728
0.9	1.777157	2.089774	0.312617

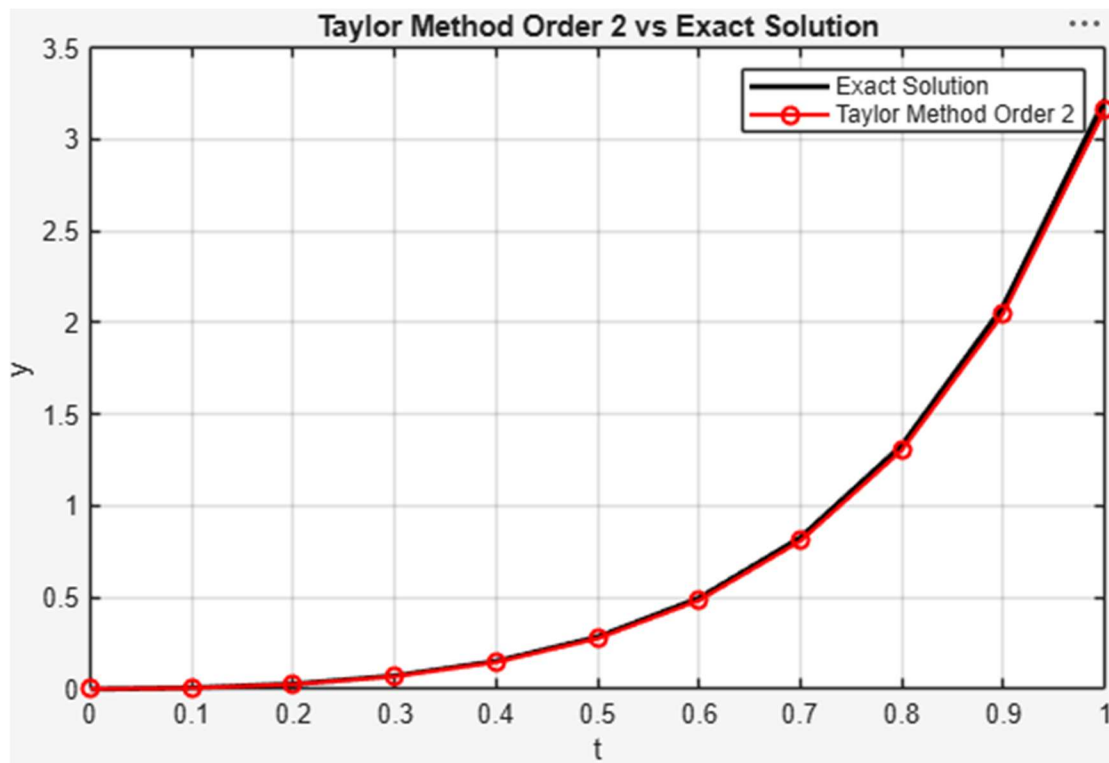


6.2 TAYLOR METHOD OF ORDER N

The Taylor Method Order 2 and 3 generated a significantly more accurate solution due to the inclusion of higher-order derivative terms.

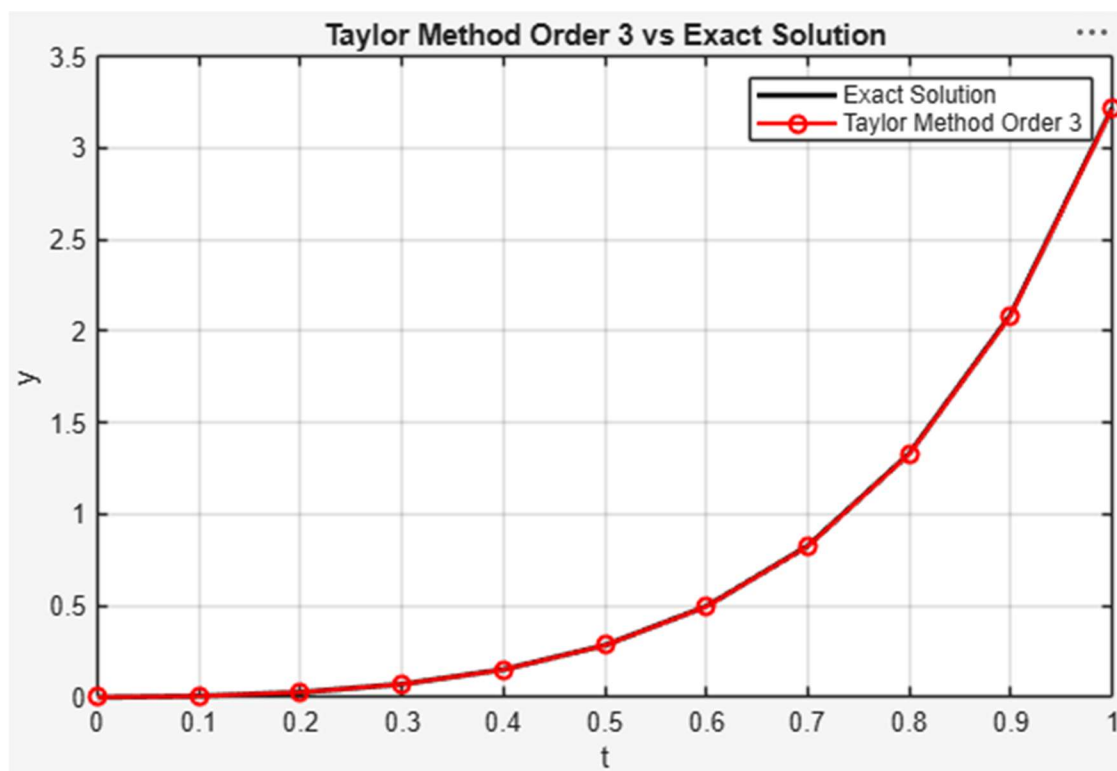
6.2.1 Order 2

t	y	y_exact	error
0.0	0.000000	0.000000	0.000000
0.1	0.005000	0.005752	0.000752
0.2	0.025023	0.026813	0.001790
0.3	0.067894	0.071145	0.003251
0.4	0.145448	0.150778	0.005329
0.5	0.275313	0.283617	0.008303
0.6	0.483454	0.496020	0.012566
0.7	0.807808	0.826481	0.018673
0.8	1.303447	1.330857	0.027410
0.9	2.049889	2.089774	0.039885



6.2.2 Order 3

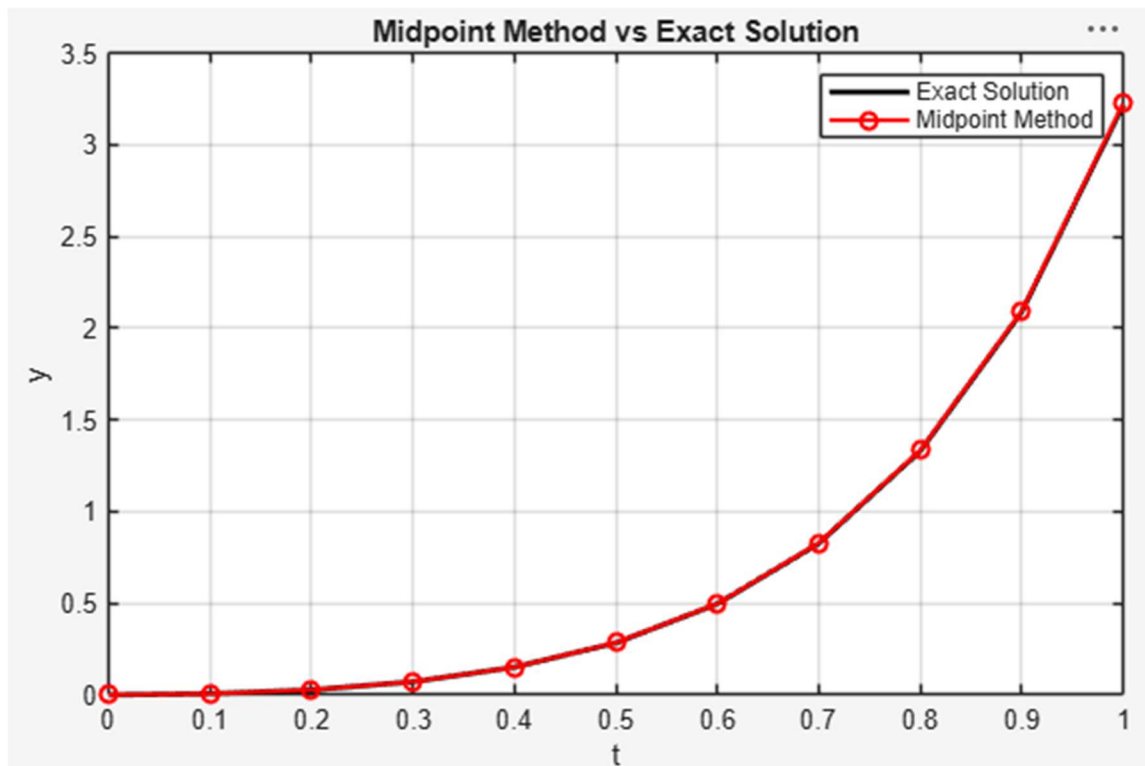
t	y	y_exact	error
0.0	0.000000	0.000000	0.000000
0.1	0.005667	0.005752	0.000085
0.2	0.026619	0.026813	0.000193
0.3	0.070807	0.071145	0.000337
0.4	0.150244	0.150778	0.000534
0.5	0.282808	0.283617	0.000809
0.6	0.494825	0.496020	0.001195
0.7	0.824740	0.826481	0.001741
0.8	1.328345	1.330857	0.002512
0.9	2.086171	2.089774	0.003603



6.3 MIDPOINT METHOD

The Midpoint Method provided a substantial improvement over the Euler Method and closely followed the exact solution.

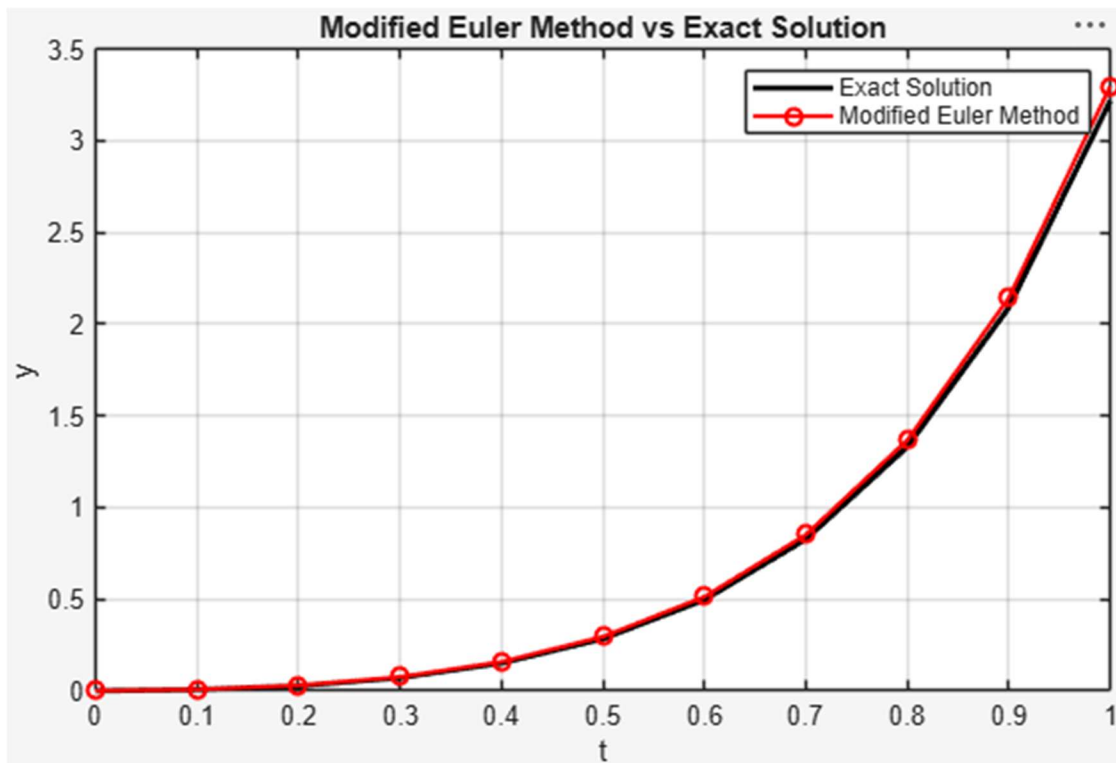
t	y	y_exact	error
0.0	0.000000	0.000000	0.000000
0.1	0.005809	0.005752	0.000057
0.2	0.026938	0.026813	0.000126
0.3	0.071370	0.071145	0.000226
0.4	0.151163	0.150778	0.000385
0.5	0.284257	0.283617	0.000640
0.6	0.497066	0.496020	0.001047
0.7	0.828161	0.826481	0.001680
0.8	1.333509	1.330857	0.002652
0.9	2.093896	2.089774	0.004121



6.4 MODIFIED EULER METHOD

The Modified Euler Method improved upon the Euler Method but produced larger errors than the Midpoint Method and Taylor Method Order 3 for the selected step size. Despite being a second-order method, its accuracy was lower than expected for this particular problem.

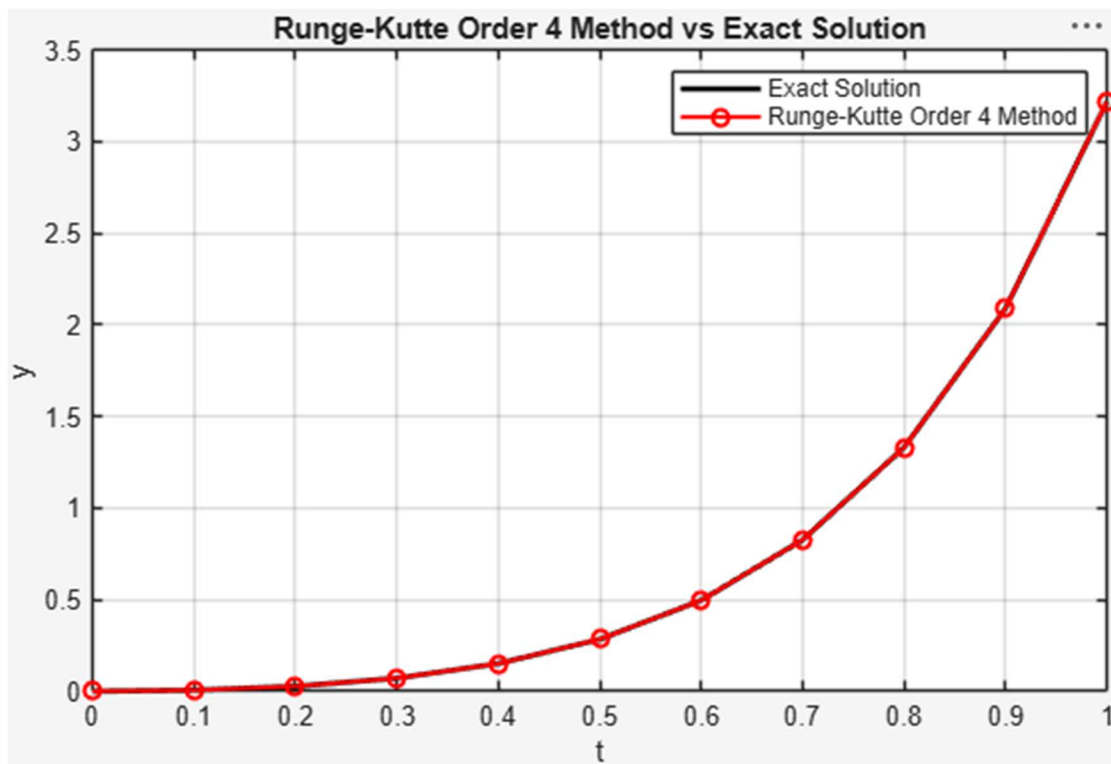
t	y	y_exact	error
0.0	0.000000	0.000000	0.000000
0.1	0.006749	0.005752	0.000997
0.2	0.029155	0.026813	0.002342
0.3	0.075378	0.071145	0.004234
0.4	0.157728	0.150778	0.006950
0.5	0.294501	0.283617	0.010884
0.6	0.512614	0.496020	0.016594
0.7	0.851351	0.826481	0.024870
0.8	1.367687	1.330857	0.036830
0.9	2.143833	2.089774	0.054059



6.5 RUNGE-KUTTA ORDER 4 METHOD

The RK4 Method produced results that were nearly identical to the exact solution throughout the interval.

t	y	y_exact	error
0.0	0.000000	0.000000	0.000000
0.1	0.005755	0.005752	0.000003
0.2	0.026819	0.026813	0.000006
0.3	0.071155	0.071145	0.000011
0.4	0.150795	0.150778	0.000017
0.5	0.283643	0.283617	0.000027
0.6	0.496060	0.496020	0.000040
0.7	0.826540	0.826481	0.000060
0.8	1.330944	1.330857	0.000087
0.9	2.089902	2.089774	0.000127



7 COMPARISON TABLE

A comparison was performed using the numerical solution at $t = 1$.

Method	Final Numerical Value	Exact Value	Absolute Error
Euler	2.7609	3.2191	0.4582
Taylor Order 3	3.2140	3.2191	0.0051
Midpoint	3.2254	3.2191	0.0063
Modified Euler	3.2979	3.2191	0.0788
RK4	3.2193	3.2191	0.0002

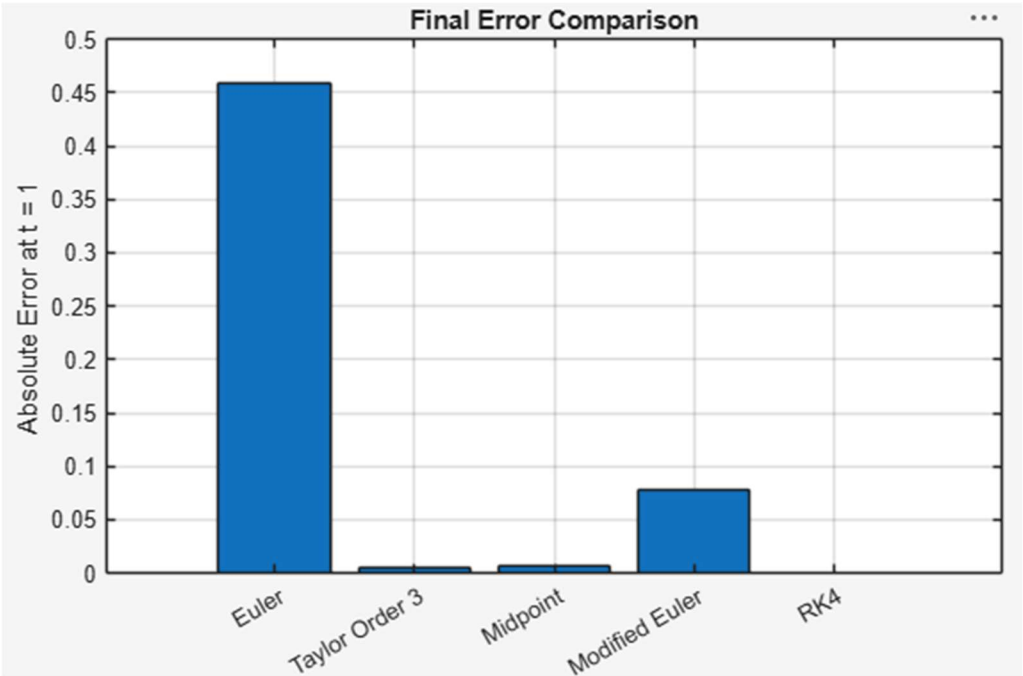
The RK4 Method achieved the smallest error, while the Euler Method produced the largest error.

7.1 RANKING

Rank	Method	Absolute Error
1	RK4	0.0002
2	Taylor Order 3	0.0051
3	Midpoint	0.0063
4	Modified Euler	0.0788
5	Euler	0.4582

8 COMPARISON GRAPH

A combined graph was generated to compare all numerical methods with the exact solution.



The graph clearly shows that all methods follow the same overall trend, but higher-order methods remain closer to the exact solution.

9 CONVERGENCE STUDY

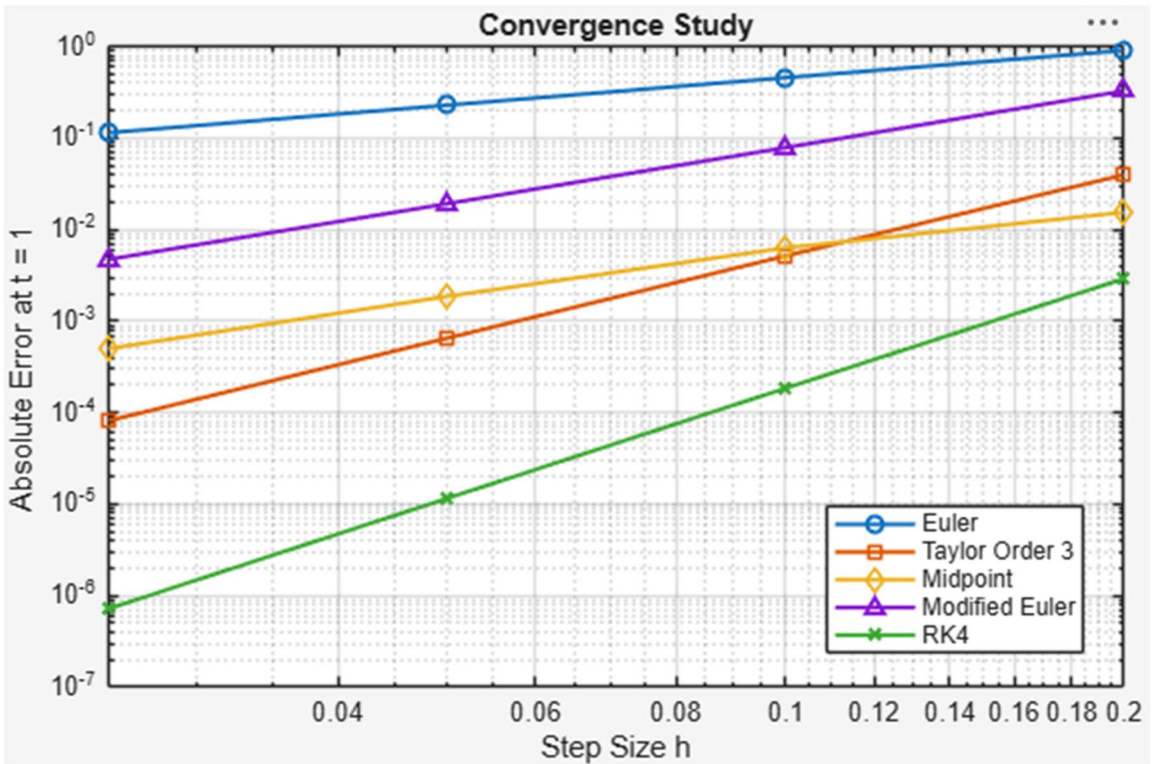
A convergence study was performed using the following step sizes:

$$h = 0.2, 0.1, 0.05, 0.025$$

For each step size, the final error at $t = 1$ was calculated.

h	Euler	Taylor_Order3	Midpoint	Modified_Euler	RK4
0.2	0.90845	0.039589	0.015461	0.32743	0.0028933
0.1	0.4582	0.0051435	0.006317	0.078791	0.00018408
0.05	0.22938	0.00065006	0.0018635	0.01921	1.1513e-05
0.025	0.11467	8.1536e-05	0.0004996	0.0047358	7.1827e-07

The results showed that the error decreased as the step size became smaller. Higher-order methods demonstrated faster convergence than lower-order methods.



The convergence plot confirms the theoretical behavior of each numerical method.

10 ERROR ANALYSIS

The observed results were consistent with numerical analysis theory.

- The Euler Method exhibited first-order accuracy and therefore had the largest error.
- The Midpoint Method and Modified Euler Method are second-order methods and produced considerably smaller errors.
- The Taylor Method Order 3 achieved higher accuracy due to the inclusion of third-order derivative information.
- The RK4 Method exhibited the highest accuracy and the fastest convergence rate among all methods considered.

11 DISCUSSION

The comparison results indicate that the RK4 Method produced the most accurate solution, with an absolute error of approximately 0.0002 at $t = 1$. This confirms the superior accuracy of fourth-order Runge-Kutta methods.

The Taylor Method Order 3 and Midpoint Method also produced highly accurate solutions, with errors below 0.01. These methods closely followed the exact solution throughout the interval and demonstrated strong convergence characteristics.

The Modified Euler Method performed significantly better than the basic Euler Method but produced a larger error than expected. This may be due to the specific implementation or the behavior of the differential equation over the selected interval.

The Euler Method produced the largest error, approximately 0.4582, demonstrating the limitations of first-order numerical methods when larger step sizes are used.

12 ENGINEERING APPLICATIONS

Numerical solution techniques for differential equations are fundamental in many engineering fields, including:

- Mechanical system simulation
- Robotics and autonomous systems
- Control system design
- Heat transfer analysis
- Fluid mechanics
- Aerospace trajectory calculations
- Electrical circuit modeling

These methods are widely used in engineering software and simulation packages.

13 CONCLUSION

In this project, five numerical methods were implemented and compared for solving an initial value problem in MATLAB.

The results demonstrated that all methods successfully approximated the solution, but their accuracy varied significantly.

The Euler Method produced the largest error, while the RK4 Method produced the most accurate results. The Taylor Method Order 3, Midpoint Method, and Modified Euler Method also provided highly accurate approximations.

The convergence study confirmed the theoretical accuracy and convergence properties of the implemented methods. Overall, the project provided valuable insight into the strengths and limitations of different numerical techniques for solving ordinary differential equations.

Among all methods considered, the Runge-Kutta Fourth Order Method (RK4) provided the highest accuracy and best overall performance. The project demonstrated how higher-order numerical methods can significantly reduce approximation errors while maintaining computational efficiency, making them valuable tools for engineering analysis and simulation.

14 FUTURE IMPROVEMENTS

Possible future extensions of this project include:

- Adaptive step-size methods
- Higher-order Taylor methods
- Systems of differential equations
- Comparison with MATLAB built-in solvers such as ode45
- Development of a graphical user interface
- Investigation of stiff differential equations

These additions would further enhance the scope and practical relevance of the project.